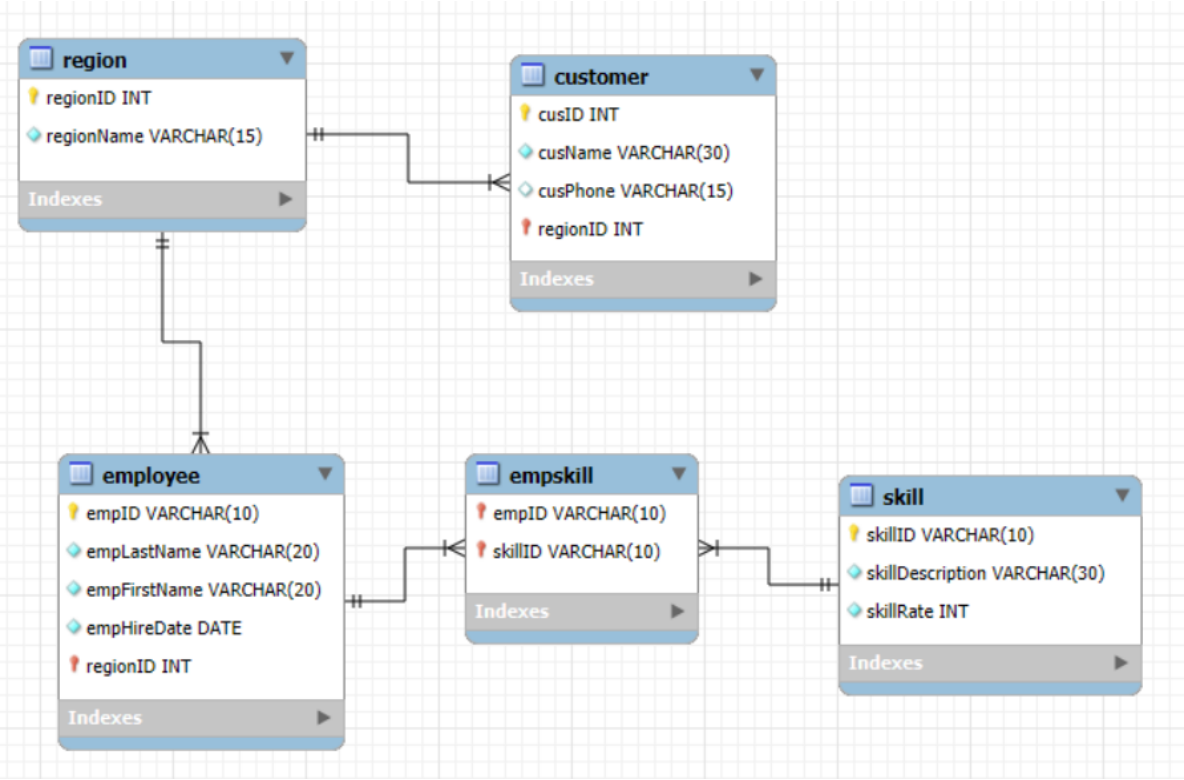
Francisco Acosta
Course Number: CEIS236
Final Project

DeVry University
College of Engineering and Information Sciences

Problem 1



Problem 2

```
DROP TABLE IF EXISTS empskill;
DROP TABLE IF EXISTS employee;
DROP TABLE IF EXISTS customer;
DROP TABLE IF EXISTS skill;
DROP TABLE IF EXISTS region;

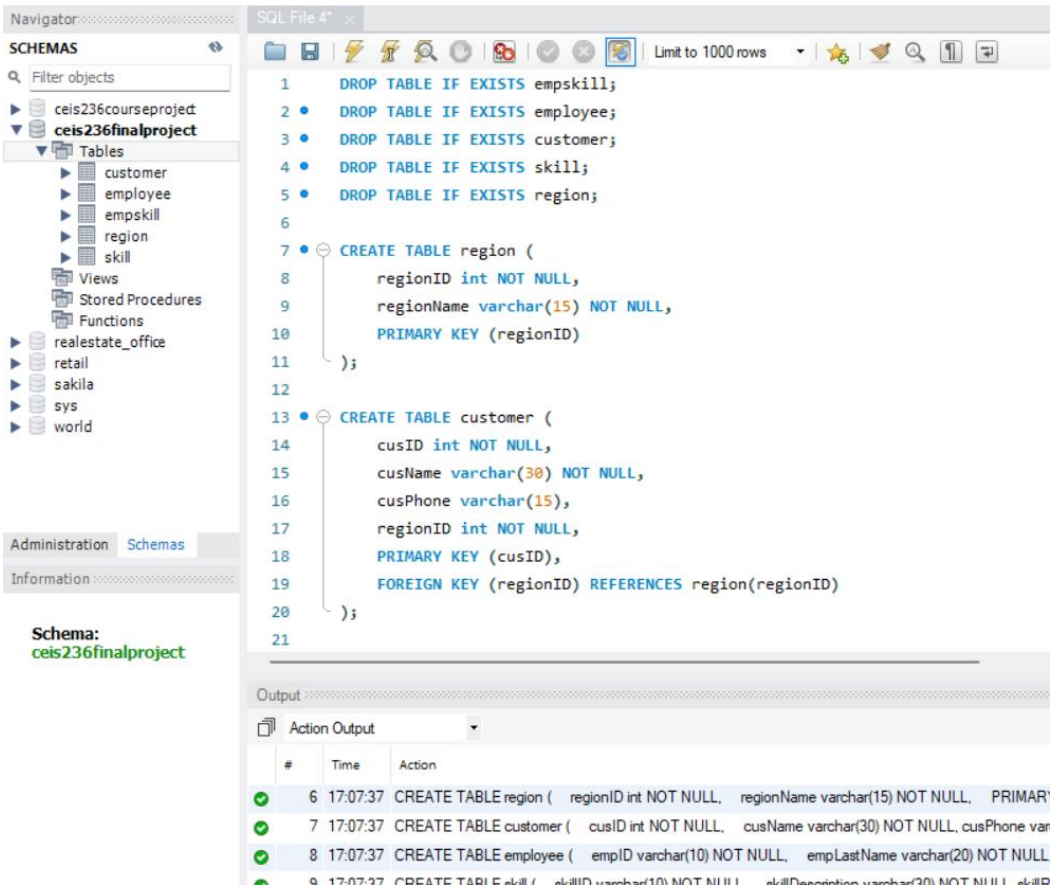
CREATE TABLE region (
   regionID int NOT NULL,
   regionName varchar(15) NOT NULL,
   PRIMARY KEY (regionID)
);

CREATE TABLE customer (
   cusID int NOT NULL,
   cusName varchar(30) NOT NULL,
         cusPhone varchar(15),
         regionID int NOT NULL,
   PRIMARY KEY (cusID),
   FOREIGN KEY (regionID) REFERENCES region(regionID)
);

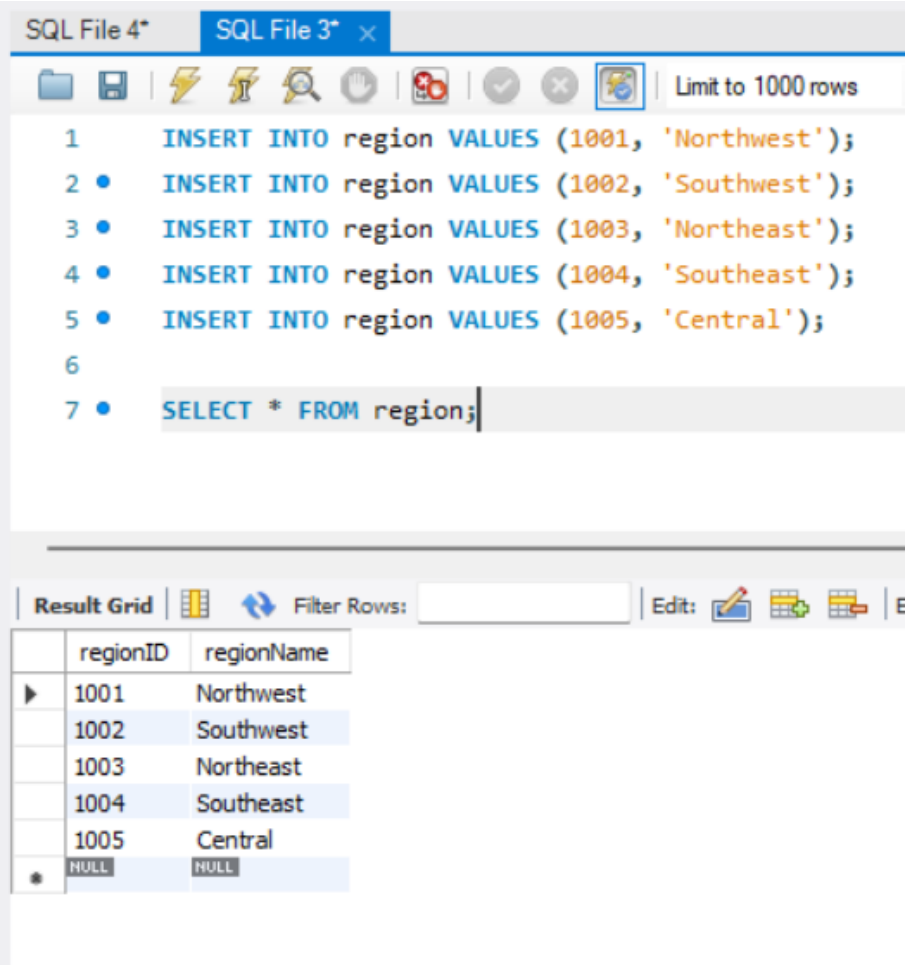
CREATE TABLE employee (
   empID varchar(10) NOT NULL,
   empLastName varchar(20) NOT NULL,
   empFirstName varchar(20) NOT NULL,
         empHireDate DATE NOT NULL,
         regionID int NOT NULL,
   PRIMARY KEY (empID),
   FOREIGN KEY (regionID) REFERENCES region(regionID)
);

CREATE TABLE skill (
   skillID varchar(10) NOT NULL,
   skillDescription varchar(30) NOT NULL,
         skillRate DOUBLE NOT NULL,
         PRIMARY KEY (skillID)
);

CREATE TABLE empSkill (
   empID varchar(10) NOT NULL,
   skillID varchar(10) NOT NULL,
         PRIMARY KEY (empID, skillID),
   FOREIGN KEY (empID) REFERENCES employee(empID),
   FOREIGN KEY (skillID) REFERENCES skill(skillID)
);
```


Navigator
SCHEMAS
Filter objects
ceis236courseproject
ceis236finalproject
Tables
customer
employee
empskill
region
skill
Views
Stored Procedures
Functions
realestate_office
retail
sakila
sys
world
Administration
Schemas
Information
Schema:
ceis236finalproject
SQL File 4*
Limit to 1000 rows
1 DROP TABLE IF EXISTS empskill;
2 DROP TABLE IF EXISTS employee;
3 DROP TABLE IF EXISTS customer;
4 DROP TABLE IF EXISTS skill;
5 DROP TABLE IF EXISTS region;
6
7 CREATE TABLE region (
8 regionID int NOT NULL,
9 regionName varchar(15) NOT NULL,
10 PRIMARY KEY (regionID)
11);
12
13 CREATE TABLE customer (
14 cusID int NOT NULL,
15 cusName varchar(30) NOT NULL,
16 cusPhone varchar(15),
17 regionID int NOT NULL,
18 PRIMARY KEY (cusID),
19 FOREIGN KEY (regionID) REFERENCES region(regionID)
20);
21
Output
Action Output
Time Action
6 17:07:37 CREATE TABLE region (regionID int NOT NULL, regionName varchar(15) NOT NULL, PRIMAR
7 17:07:37 CREATE TABLE customer (cusID int NOT NULL, cusName varchar(30) NOT NULL, cusPhone var
8 17:07:37 CREATE TABLE employee (empID varchar(10) NOT NULL, empLastName varchar(20) NOT NULL

Problem 3


SQL File 4*
SQL File 3*
Limit to 1000 rows
1 INSERT INTO region VALUES (1001, 'Northwest');
2 INSERT INTO region VALUES (1002, 'Southwest');
3 INSERT INTO region VALUES (1003, 'Northeast');
4 INSERT INTO region VALUES (1004, 'Southeast');
5 INSERT INTO region VALUES (1005, 'Central');
6
7 SELECT * FROM region;
Result Grid
Filter Rows:
Edit:
regionID regionName
1001 Northwest
1002 Southwest
1003 Northeast
1004 Southeast
1005 Central
NULL NULL

Problem 4

```
1    INSERT INTO customer VALUES(1, 'Bellsouth', '222-333-4571', 1004);
2    INSERT INTO customer VALUES(2, 'Comcast', '253-444-5555', 1003);
3    INSERT INTO customer VALUES(3, 'Ford', '367-555-6666', 1005);
4    INSERT INTO customer VALUES(4,'Exxon', '444-777-7777', 1004);
5    INSERT INTO customer VALUES(5, 'Tesla', '555-222-8888', 1005);
6
7    SELECT * FROM customer;
```

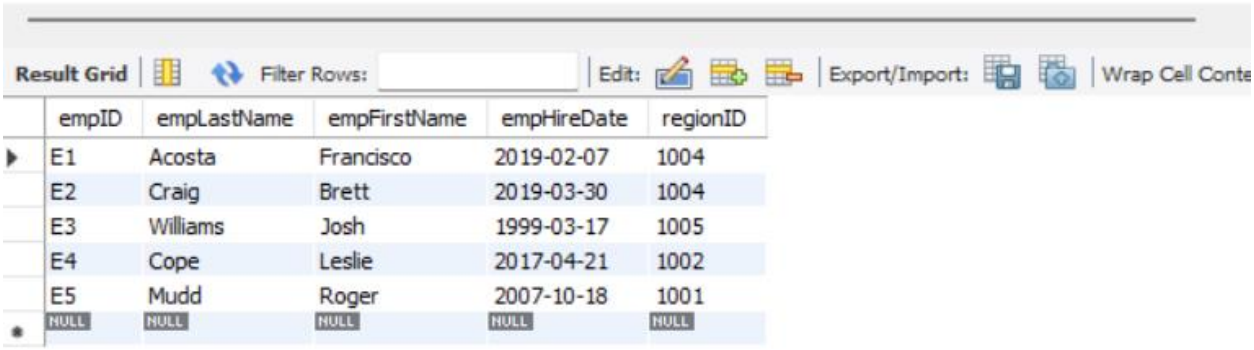
Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap

cusID	cusName	cusPhone	regionID
1	Bellsouth	222-333-4571	1004
2	Comcast	253-444-5555	1003
3	Ford	367-555-6666	1005
4	Exxon	444-777-7777	1004
5	Tesla	555-222-8888	1005
NULL	NULL	NULL	NULL

Problem 5



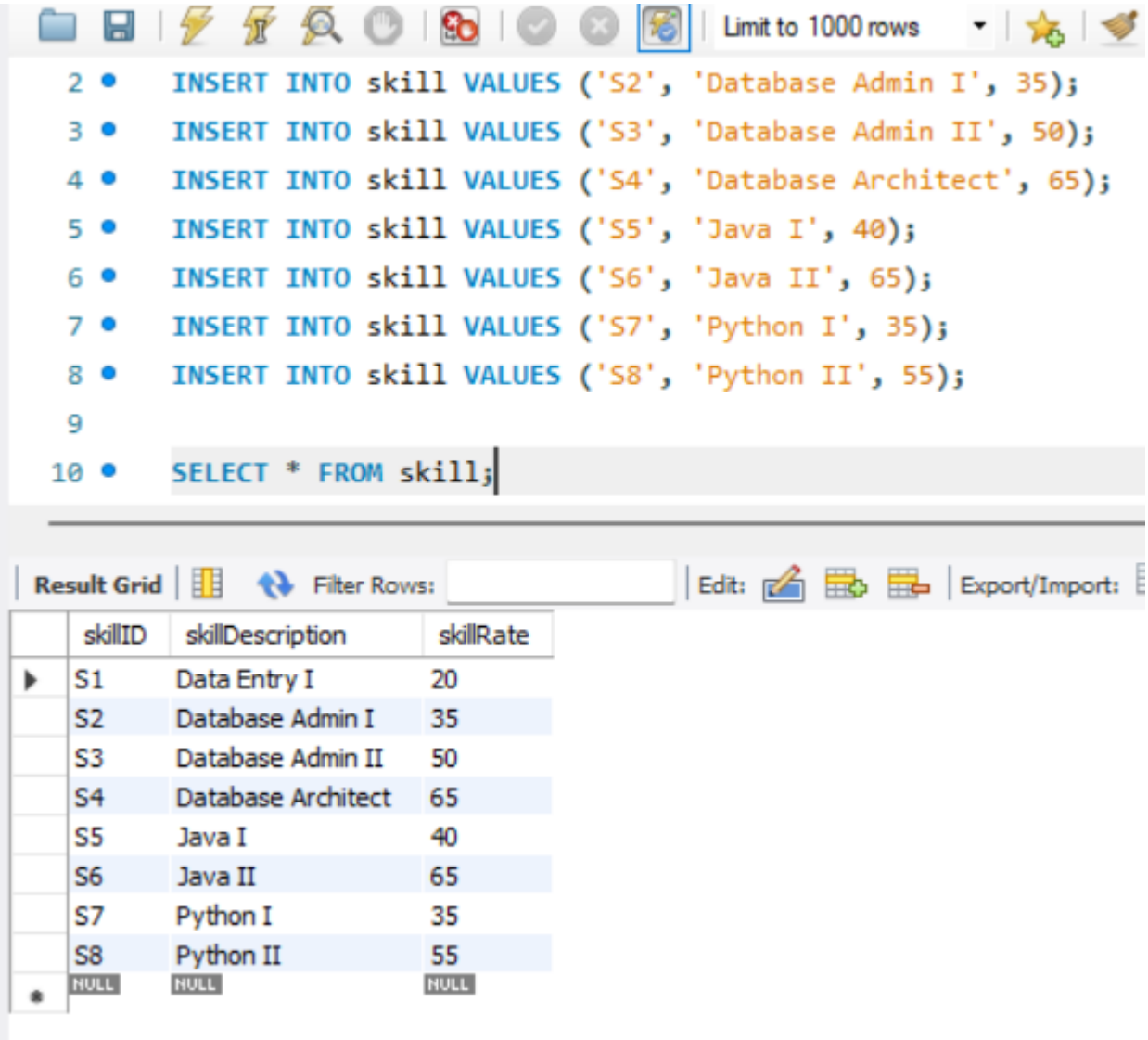
```
1  INSERT INTO employee VALUES ('E1', 'Acosta', 'Francisco', '2019-2-7', 1004);
2  INSERT INTO employee VALUES ('E2', 'Craig', 'Brett', '2019-3-30', 1004);
3  INSERT INTO employee VALUES ('E3', 'Williams', 'Josh', '1999-3-17', 1005);
4  INSERT INTO employee VALUES ('E4', 'Cope', 'Leslie', '2017-4-21', 1002);
5  INSERT INTO employee VALUES ('E5', 'Mudd', 'Roger', '2007-10-18', 1001);
6
7  SELECT * FROM employee;
```



Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Conte

empID	empLastName	empFirstName	empHireDate	regionID
E1	Acosta	Francisco	2019-02-07	1004
E2	Craig	Brett	2019-03-30	1004
E3	Williams	Josh	1999-03-17	1005
E4	Cope	Leslie	2017-04-21	1002
E5	Mudd	Roger	2007-10-18	1001
NULL	NULL	NULL	NULL	NULL

Problem 6

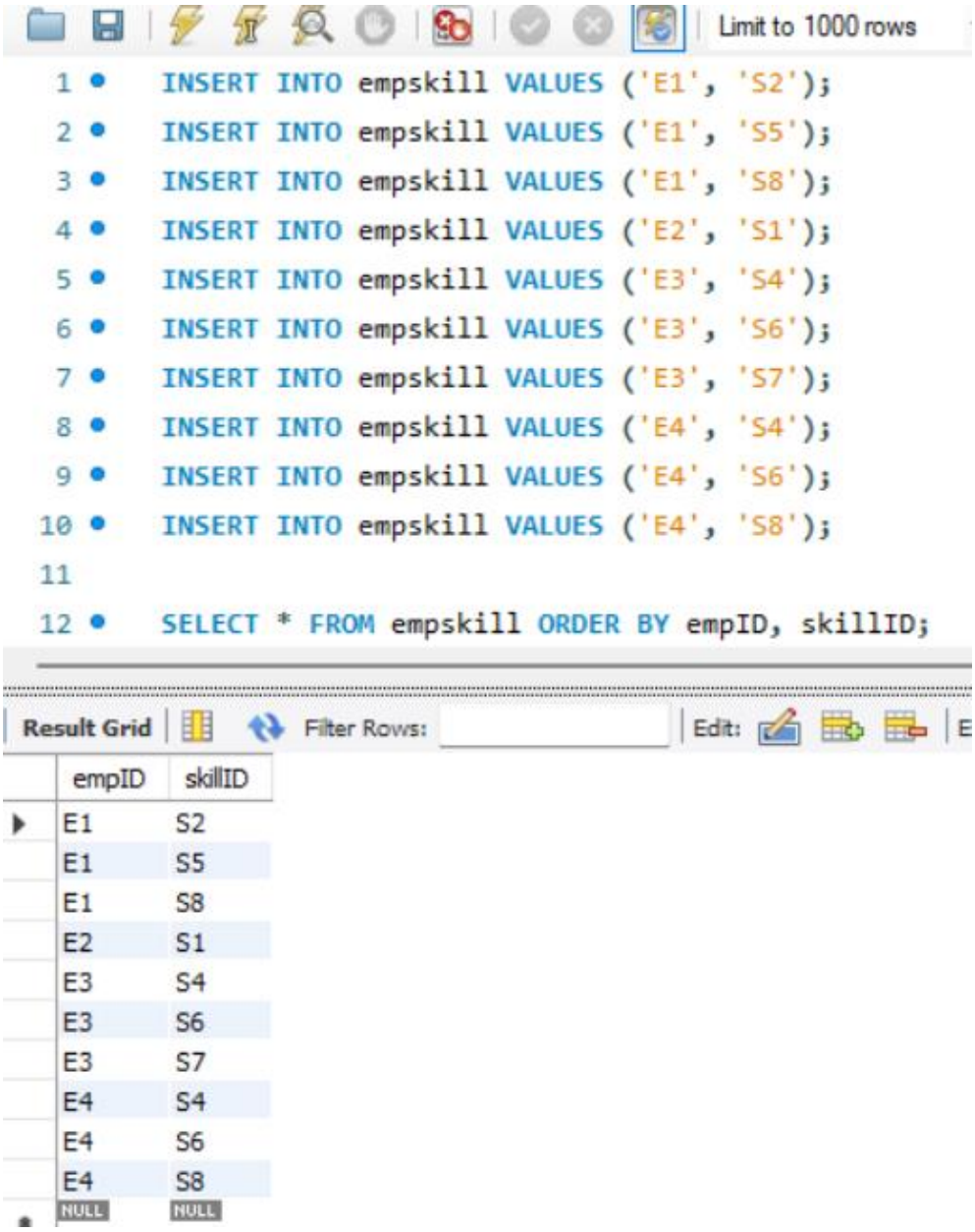


```
2  INSERT INTO skill VALUES ('S2', 'Database Admin I', 35);
3  INSERT INTO skill VALUES ('S3', 'Database Admin II', 50);
4  INSERT INTO skill VALUES ('S4', 'Database Architect', 65);
5  INSERT INTO skill VALUES ('S5', 'Java I', 40);
6  INSERT INTO skill VALUES ('S6', 'Java II', 65);
7  INSERT INTO skill VALUES ('S7', 'Python I', 35);
8  INSERT INTO skill VALUES ('S8', 'Python II', 55);
9
10 SELECT * FROM skill;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

skillID	skillDescription	skillRate
S1	Data Entry I	20
S2	Database Admin I	35
S3	Database Admin II	50
S4	Database Architect	65
S5	Java I	40
S6	Java II	65
S7	Python I	35
S8	Python II	55
NULL	NULL	NULL

Problem 7


Limit to 1000 rows
1 • INSERT INTO empskill VALUES ('E1', 'S2');
2 • INSERT INTO empskill VALUES ('E1', 'S5');
3 • INSERT INTO empskill VALUES ('E1', 'S8');
4 • INSERT INTO empskill VALUES ('E2', 'S1');
5 • INSERT INTO empskill VALUES ('E3', 'S4');
6 • INSERT INTO empskill VALUES ('E3', 'S6');
7 • INSERT INTO empskill VALUES ('E3', 'S7');
8 • INSERT INTO empskill VALUES ('E4', 'S4');
9 • INSERT INTO empskill VALUES ('E4', 'S6');
10 • INSERT INTO empskill VALUES ('E4', 'S8');
11
12 • SELECT * FROM empskill ORDER BY empID, skillID;
Result Grid
Filter Rows:
Edit:
empID skillID
E1 S2
E1 S5
E1 S8
E2 S1
E3 S4
E3 S6
E3 S7
E4 S4
E4 S6
E4 S8
NULL NULL

Problem 8

```
1 •  SELECT ROUND(AVG(skillRate), 2) AS Average,
2    MAX(skillRate) As Maximum, MIN(skillRate) AS Minimum
3    FROM skill;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Average	Maximum	Minimum
45.62	65	20

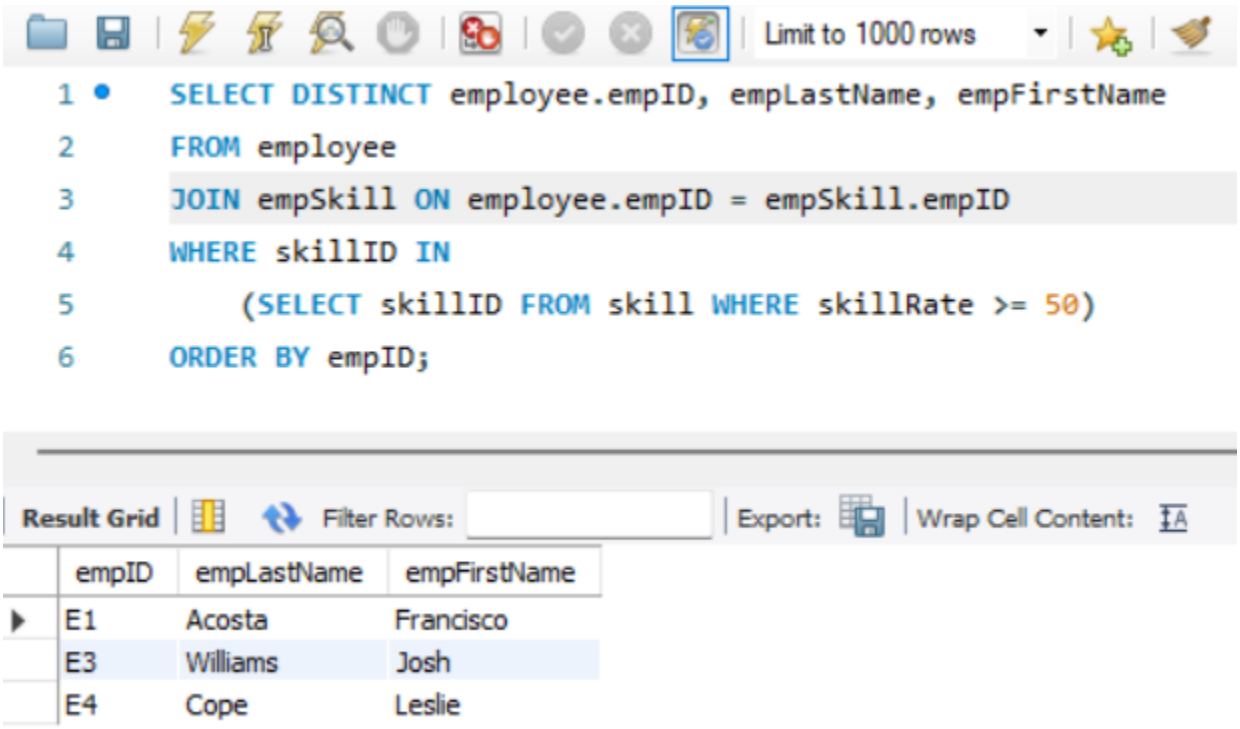
Problem 9

```
1 •  SELECT cusName
2    FROM customer
3    JOIN region ON customer.regionID = region.regionID
4    WHERE regionName = 'Central';
```

Result Grid | Filter Rows: | Export: | Wrap Cell C

cusName
Ford
Tesla

Problem 10

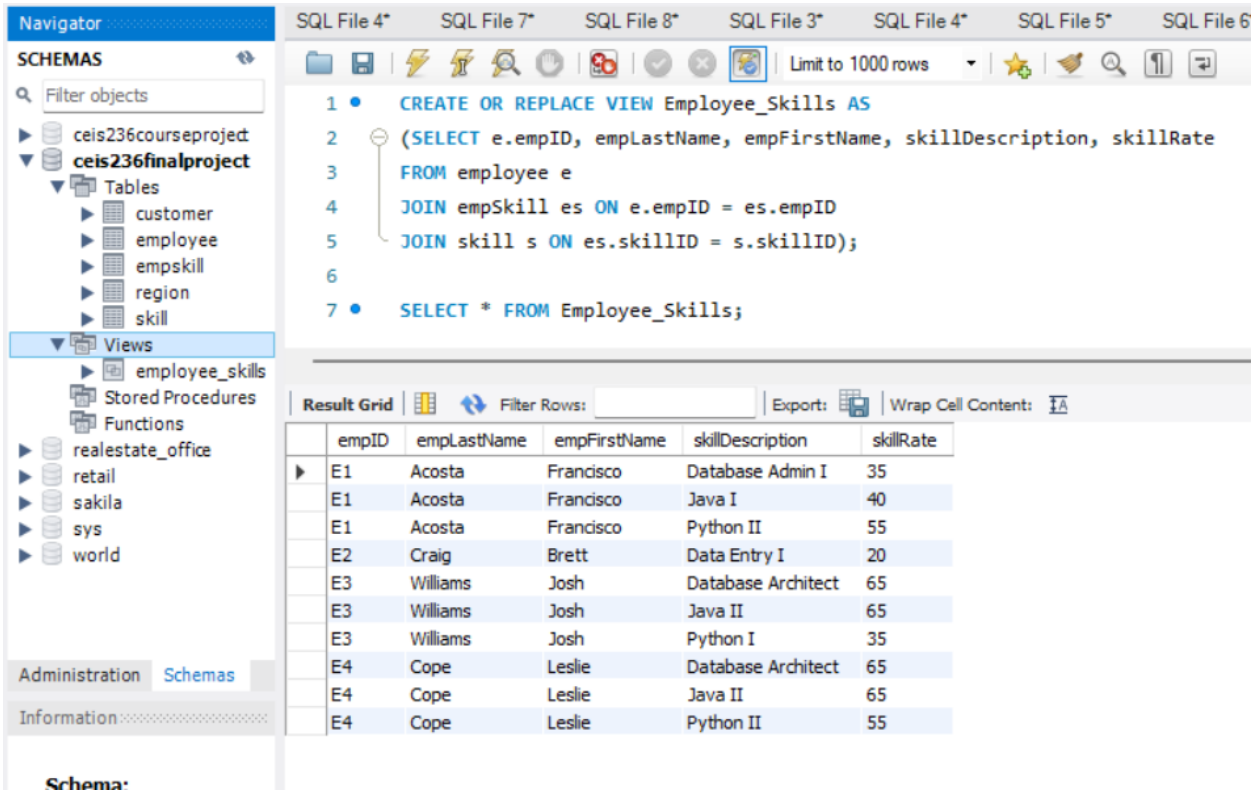


```
1  SELECT DISTINCT employee.empID, empLastName, empFirstName
2  FROM employee
3  JOIN empSkill ON employee.empID = empSkill.empID
4  WHERE skillID IN
5      (SELECT skillID FROM skill WHERE skillRate >= 50)
6  ORDER BY empID;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

empID	empLastName	empFirstName
E1	Acosta	Francisco
E3	Williams	Josh
E4	Cope	Leslie

Problem 11



empID	empLastName	empFirstName	skillDescription	skillRate
E1	Acosta	Francisco	Database Admin I	35
E1	Acosta	Francisco	Java I	40
E1	Acosta	Francisco	Python II	55
E2	Craig	Brett	Data Entry I	20
E3	Williams	Josh	Database Architect	65
E3	Williams	Josh	Java II	65
E3	Williams	Josh	Python I	35
E4	Cope	Leslie	Database Architect	65
E4	Cope	Leslie	Java II	65
E4	Cope	Leslie	Python II	55